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2. For the Web application automate the continuous integration and deployment process using Jenkins.
🕐 Write the code in Visual Studio and push it to your GITHUB repository.
🕐 Modify the code and commit the updates back in GITHUB.
🕐 Using Jenkins configure the job as pipeline and create the stages for the Continuous Integration process.
🕐 Checkout GITHUB
🕐 Build using Maven
🕐 Create Docker image
🕐 Push to Docker Hub
🕐 Create Cluster in Kubernetec and Deploy

e. The course registration details of a student.

CREATED GIT HUB :
CODES:JAVA APP CODE:

```
package com.example;

import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.GetMapping;

@RestController
@RequestMapping("/student")
public class StudentController {

  @GetMapping("/details")
  public String getDetails() {
    return "Student Name: John Doe, Course: Computer Science";
  }
}
```

DEPLOYMENT.YAML:

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: student-app
spec:
 replicas: 2
 selector:
  matchLabels:
   app: student-app
 template:
  metadata:
   labels:
    app: student-app
  spec:
   containers:
```

```yaml
    - name: student-app
      image: YOUR_DOCKERHUB_USERNAME/student-reg-app:latest
      ports:
      - containerPort: 8080
---
apiVersion: v1
kind: Service
metadata:
  name: student-service
spec:
  type: LoadBalancer
  ports:
  - port: 80
    targetPort: 8080
  selector:
    app: student-app
```

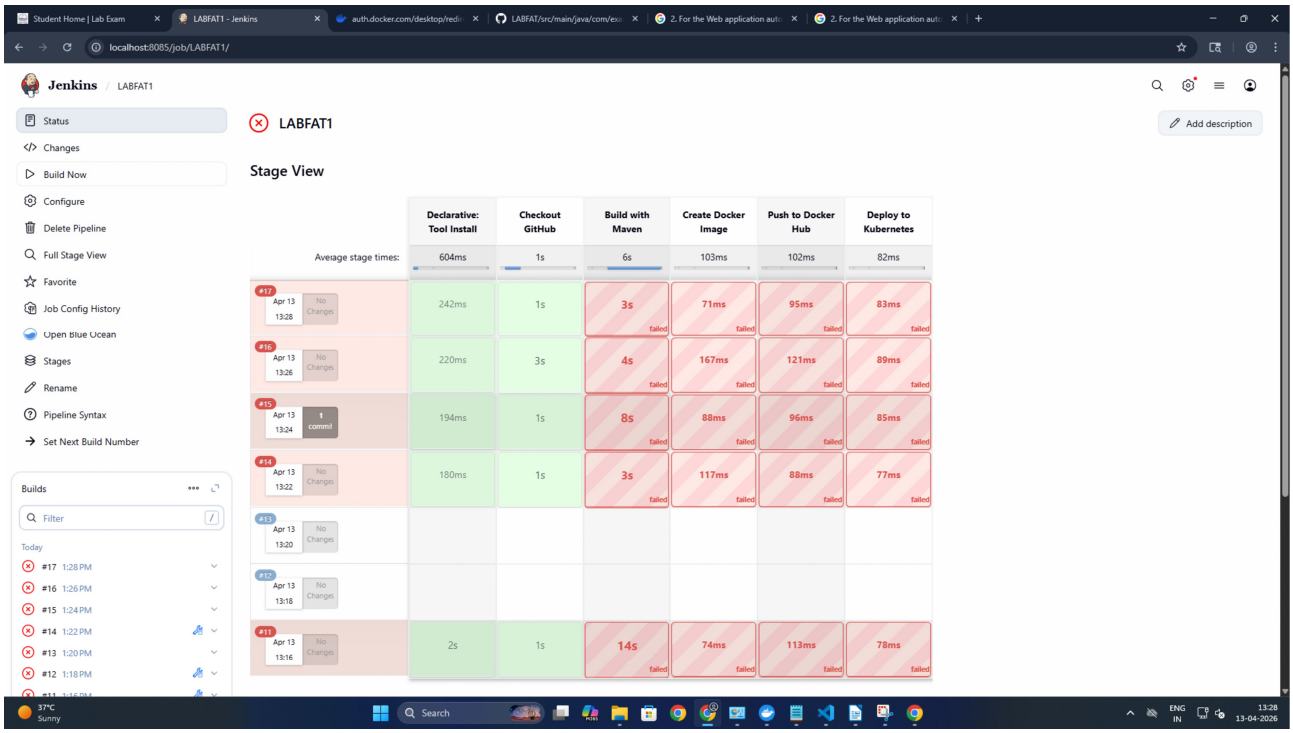
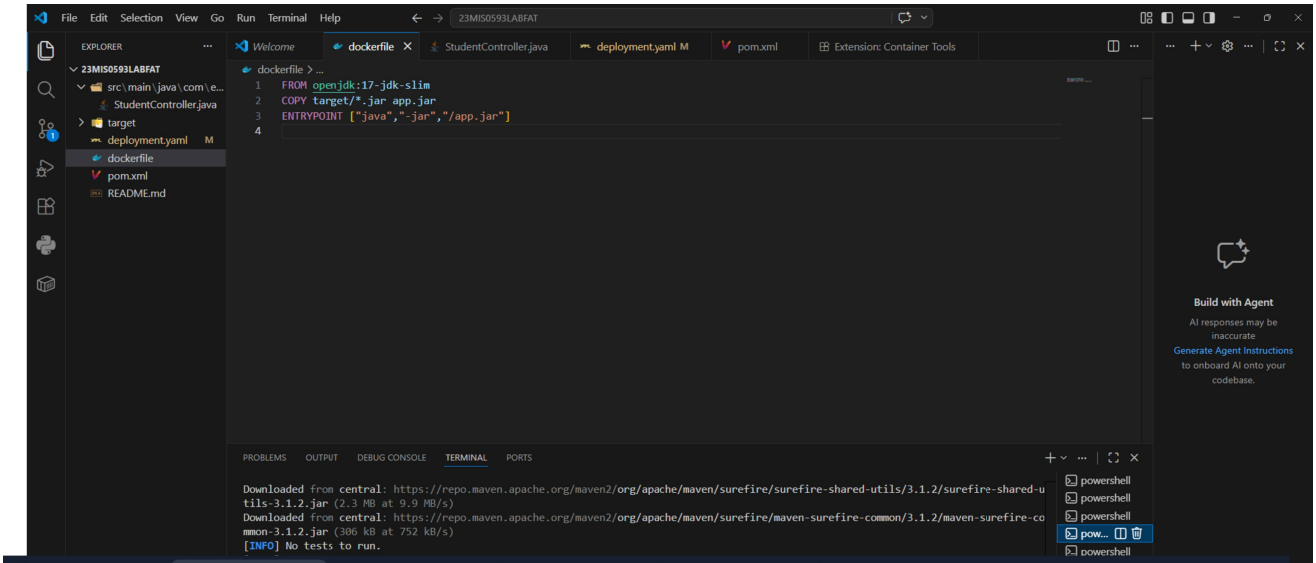
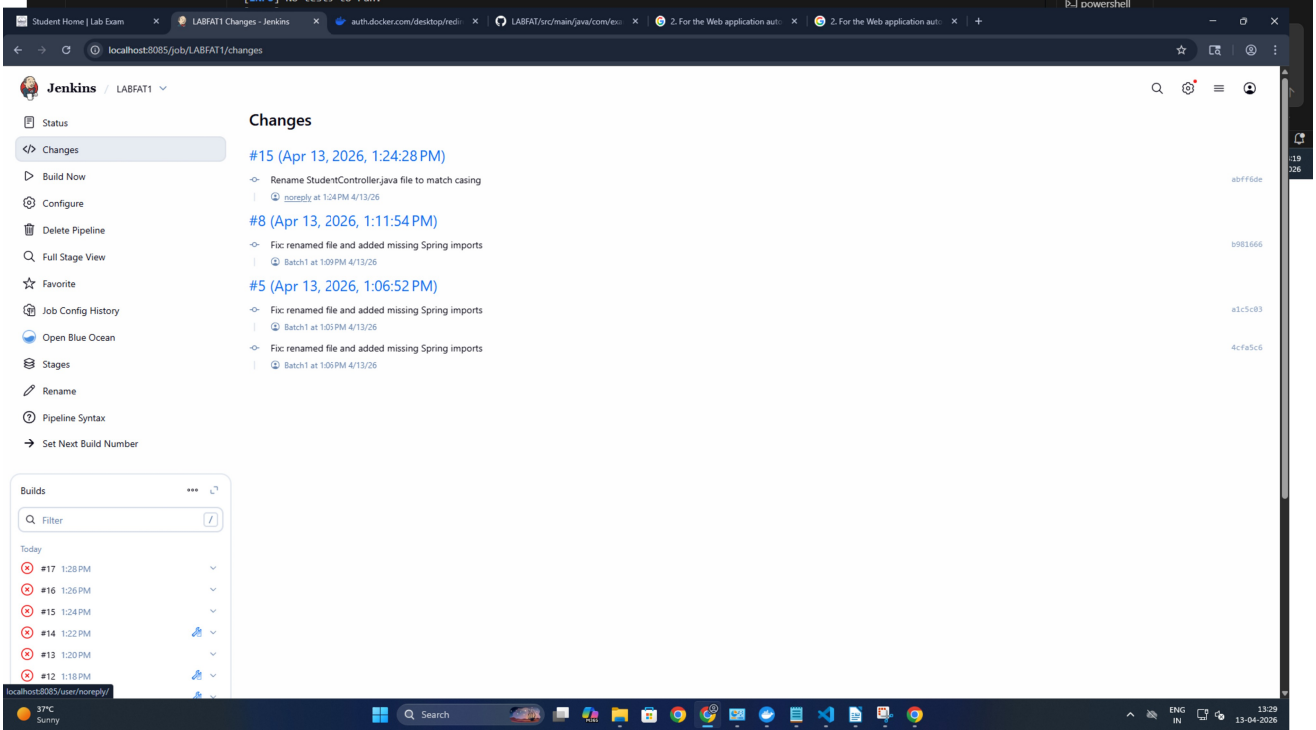
POM.XML:

```xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://apache.org"
     xmlns:xsi="http://w3.org"
     xsi:schemaLocation="http://apache.org https://apache.org">
  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.2.0</version>
    <relativePath/>
  </parent>

  <groupId>com.student</groupId>
  <artifactId>registration-app</artifactId>
  <version>1.0.0</version>

  <properties>
    <java.version>17</java.version>
  </properties>

  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-test</artifactId>
      <scope>test</scope>
    </dependency>
  </dependencies>
```

```
  <build>
    <plugins>
      <plugin>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-maven-plugin</artifactId>
      </plugin>
    </plugins>
    <finalName>student-app</finalName>
  </build>
</project>
```

DOCKERFILES:

```
FROM openjdk:17-jdk-slim
COPY target/*.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "/app.jar"]
```

PIPELINE:

screenshots: